

IBM Cognos Analytics - Case Study Scenarios

Case Study 6

SwiftMotors - Vehicle Manufacturing & Quality Analytics

Industry: Automobile Manufacturing

Modules: Reporting, Exploration, Dashboard, Story Making

Real-Time Business Scenario

SwiftMotors is an Indian automobile manufacturer producing sedans, SUVs, and electric vehicles at 3 plants — Pune, Chennai, and Ahmedabad. The production head is concerned about rising defect rates, increasing production downtime, and escalating raw material costs. The Quality Control team wants to use IBM Cognos Analytics to monitor production efficiency, defect patterns, and supplier performance to achieve Six Sigma quality standards.

Dataset / Data Available

- Vehicle ID, Model Name, Vehicle Type (Sedan/SUV/EV), Plant Location, Production Date
- Shift (Morning/Afternoon/Night), Assembly Line, Supervisor Name, Workers Count
- Units Produced, Units Defective, Defect Type, Rework Required (Yes/No), Scrap Units
- Raw Material Supplier, Material Cost per Unit, Production Time (hours), Downtime (minutes)

Case Study Questions

- Q1** **[Reporting]** Create a Production Quality Report grouped by Plant Location showing Total Units Produced, Total Defective Units, and Defect Rate %. Add a calculated column for Defect Rate = $(\text{Defective} / \text{Produced}) * 100$. Apply Conditional Styling: Red for Defect Rate > 5%, Amber for 2–5%, Green for < 2%. Add Header with plant logo text, Footer with page number. Include prompts for Plant and Vehicle Type.
- Q2** **[Reporting]** Build a Downtime Analysis Report listing Assembly Lines with total downtime minutes per month. Add a prompt for Plant and Shift selection. Sort by Total Downtime descending. Apply Conditional Styling: lines with downtime > 500 minutes highlighted in Red. Use Tahoma font, size 11. Summarize the total production hours lost due to downtime at the report footer.
- Q3** **[Reporting]** Using Report Visualize Prompt, create a visualization where Quality Managers can select chart type to view Defect Type Distribution across all plants. Add a second prompt for Date Range. Summarize the top 3 most frequent defect types as a bullet list text object within the report. Apply font color Navy Blue for all column headers.

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- Q4 [Exploration]** Using Cognos Exploration, explore the relationship between Shift Type (Morning/Afternoon/Night) and Defect Rate. Investigate whether specific supervisors or assembly lines are associated with higher defect rates. Also explore the correlation between Raw Material Supplier and Scrap Units. Present key findings with visualizations.
- Q5 [Dashboard]** Design a Manufacturing Operations Dashboard with: (1) KPI tiles — Total Units Produced, Defect Rate %, Downtime Hours, and Scrap Cost, (2) A Bar chart for Units Produced by Plant, (3) A Line chart for Monthly Defect Trend, (4) A Pie chart for Defect Type Distribution, (5) A table for Top 5 Suppliers by Material Cost. Add Plant and Date Range filters.
- Q6 [Story Making]** Create a Cognos Story for the SwiftMotors Board Quality Review: Scene 1 — Production Overview (All 3 Plants), Scene 2 — Defect Rate Analysis by Plant & Shift, Scene 3 — Top Defect Types & Root Cause Patterns, Scene 4 — Downtime Impact on Production Output, Scene 5 — Supplier Performance & Material Cost Analysis, Scene 6 — Quality Improvement Roadmap & Targets.
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